

Batch: C3 Roll No.: 16010123325

Experiment No. 2

Grade: AA / AB / BB / BC / CC / CD /DD

Title: Virtualization in Cloud using Oracle VirtualBox or KVM

Objective:

To understand the concept of virtualization in cloud computing by creating and managing virtual machines using Oracle VirtualBox or KVM

Expected Outcome of Experiment:

CO	Outcome
CO2	Investigate the system virtualization and outline its role in enabling the cloud computing System model

Books/ Journals/ Websites referred:

<http://www.virtualbox.org/wiki/Downloads>

Abstract:-

Virtualization is a method of separating a service from its underlying physical infrastructure. It involves creating a virtual replica of a computing resource, such as computer hardware, using specialized software. Initially developed during the era of mainframes, virtualization allows multiple operating systems and applications to run simultaneously on the same hardware. This enhances hardware utilization and flexibility by enabling the concurrent execution of different software environments.

Related Theory: -

Linux KVM

Kernel-based Virtual Machine (KVM) is a complete virtualization solution for Linux operating on x86 hardware with Intel VT or AMD-V support. It consists of:

- A processor-specific module (`kvm-intel.ko` or `kvm-amd.ko`).
- A loadable kernel module (`kvm.ko`), which provides the fundamental virtualization architecture.

KVM enables multiple virtual machines (VMs) to run unmodified Linux or Windows images. Each VM has access to private virtualized hardware, including a network card, disk, and graphics adapter. KVM is open-source software, and since Linux kernel version 2.6.20, its kernel component has been integrated into the mainline Linux kernel. As of version 1.3, its user-space component is part of mainstream QEMU.

Oracle VirtualBox

Oracle VM VirtualBox is a type-2 hypervisor for x86 virtualization, developed by Oracle Corporation. It was originally created by InnoTek Systemberatung GmbH and later acquired by Sun Microsystems in 2008, which was then purchased by Oracle in 2010.

VirtualBox supports multiple host operating systems, including Microsoft Windows, macOS, Linux, Solaris, and OpenSolaris. It also has ports for FreeBSD and Genode. VirtualBox allows the creation and management of guest virtual machines running various operating systems such as Windows, Linux, BSD, OS/2, Solaris, Haiku, and OSx86.

A "Guest Additions" package is available for certain guest operating systems, improving performance—especially graphics—and enabling dynamic resolution adjustment when resizing the virtual machine window.

Virtualization

VirtualBox allows users to run multiple guest operating systems alongside a single host OS. Each guest OS operates within its own VM, which can be independently started, paused, and stopped. Depending on the underlying host hardware, users can configure each VM for either software-based or hardware-assisted virtualization.

Key features include:

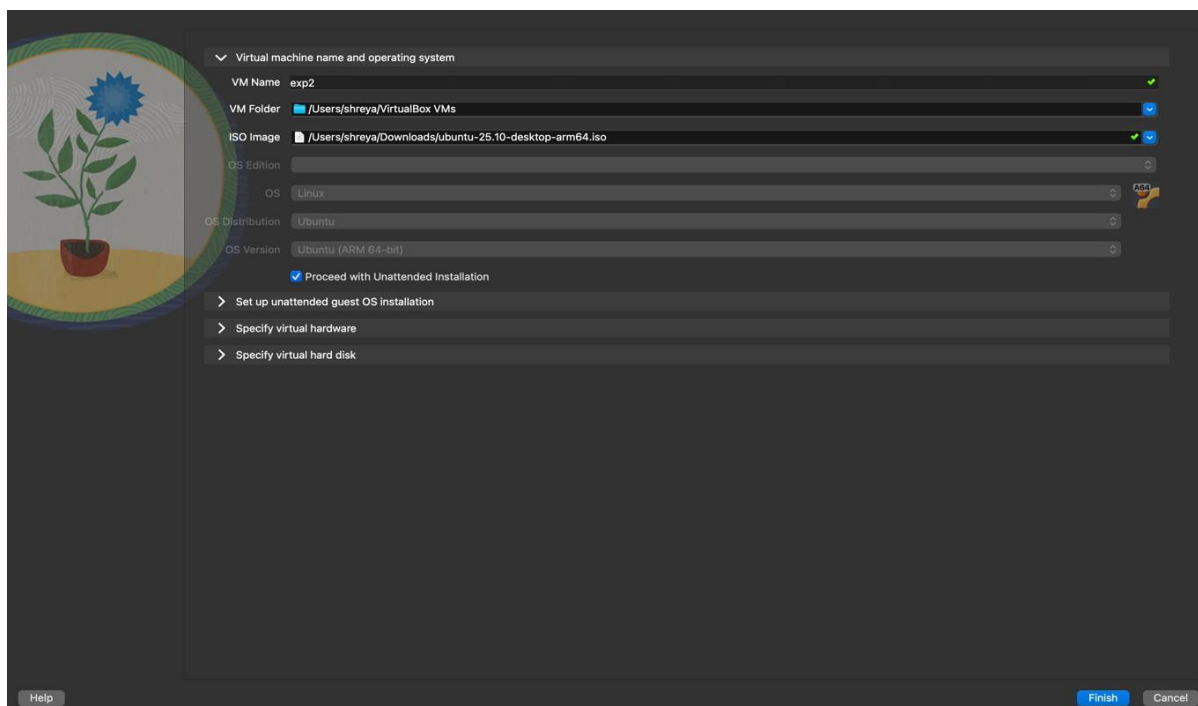
- A shared clipboard for communication between host and guest OS.
- A virtualized network facility for seamless connectivity.
- Direct communication between guest VMs if properly configured.

Virtualization enhances system efficiency, optimizes resource usage, and provides a flexible computing environment.

Implementation Details:

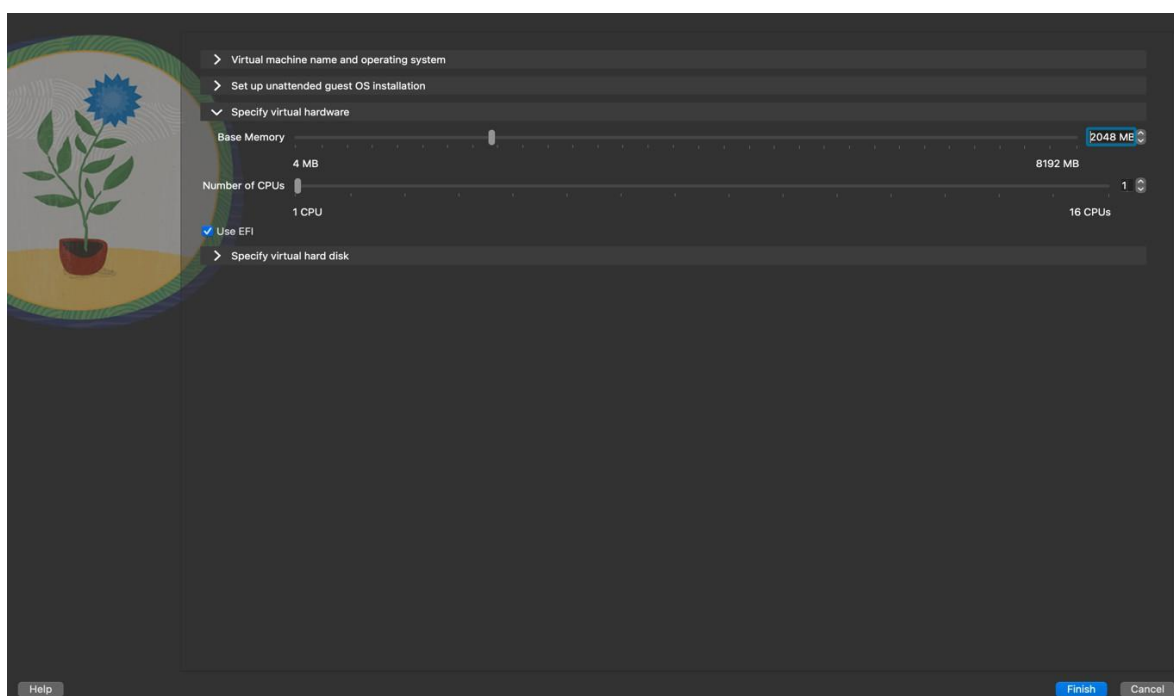
Create a New Virtual Machine

1. Enter Name (e.g., Linux_VM).
2. Select Type as Linux.



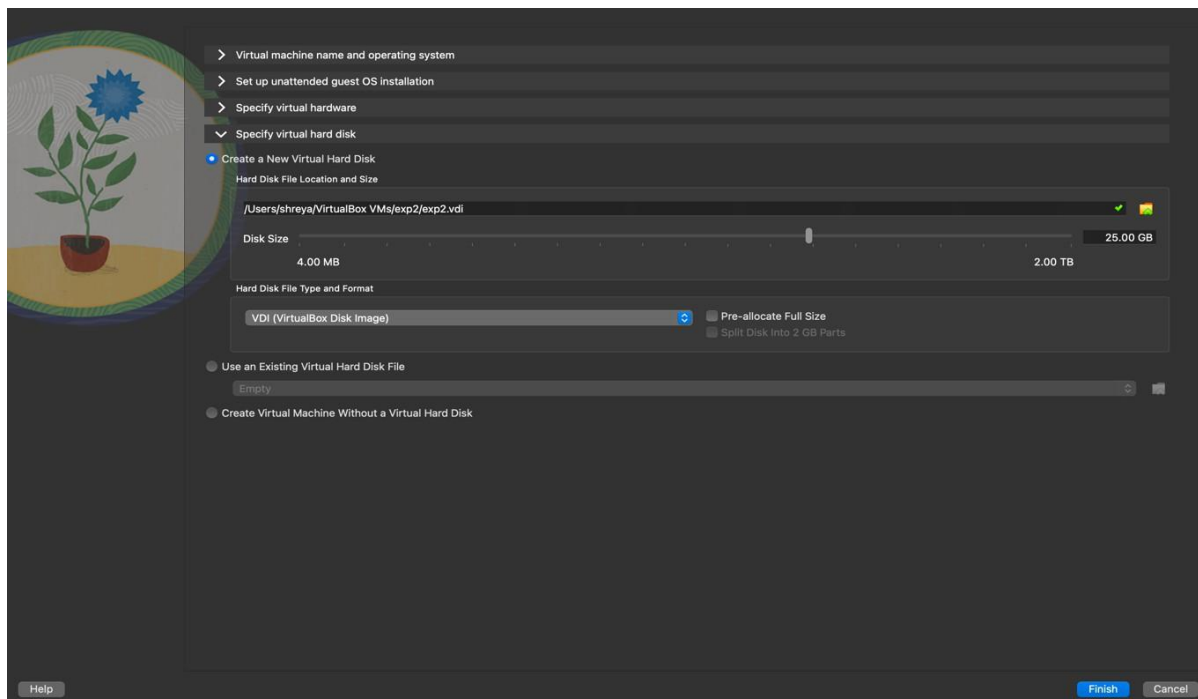
Allocate Memory (RAM)

1. Allocate at least 2048 MB RAM.

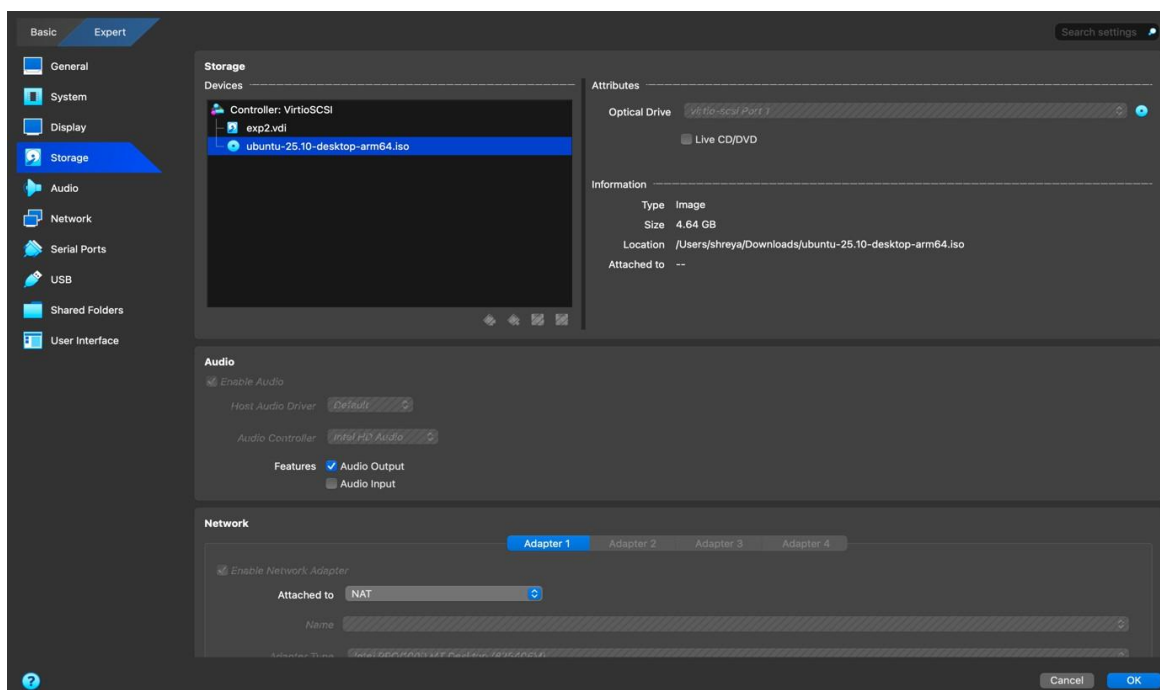


Create Virtual Hard Disk

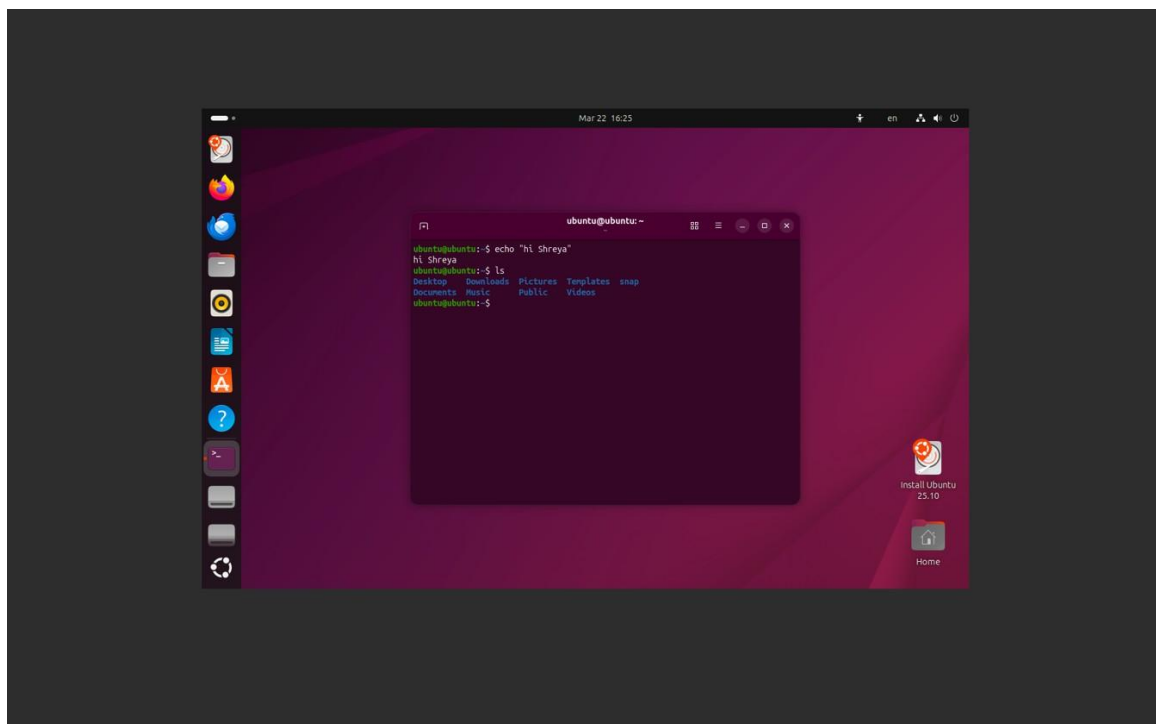
1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image).
3. Set disk size to 25 GB.



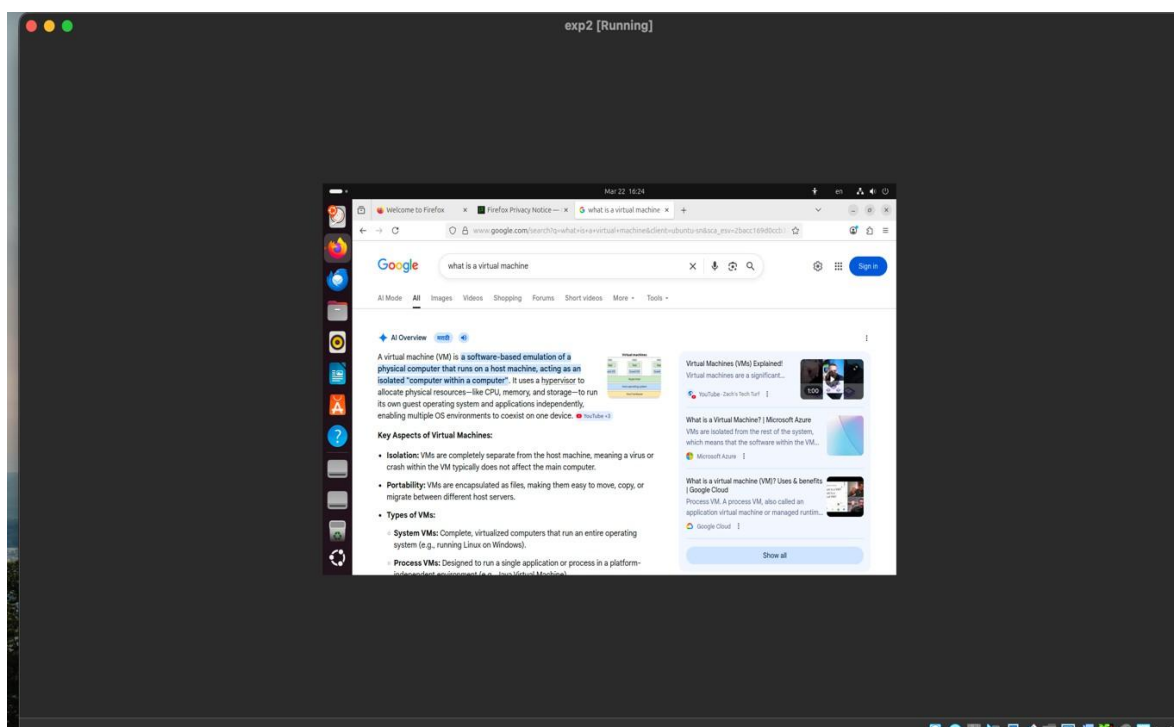
Attach Linux ISO File



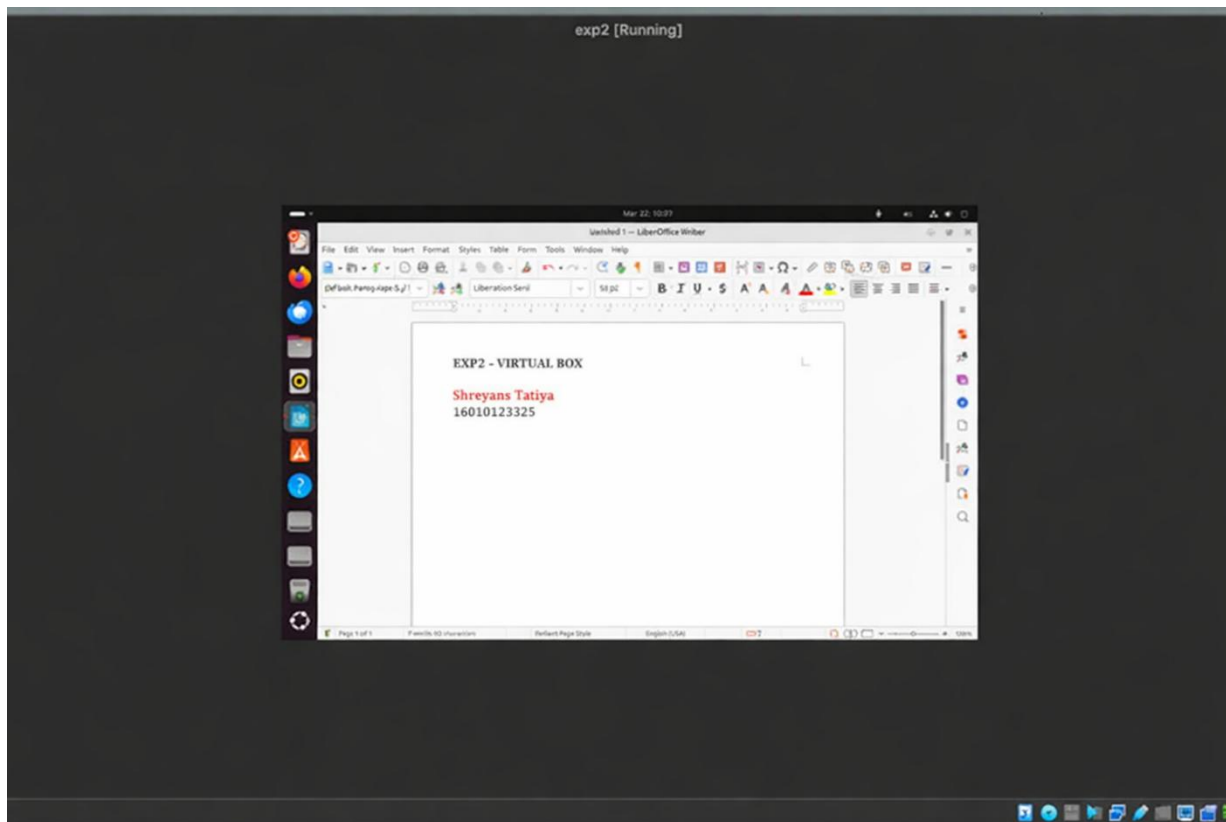
Using Terminal



Firefox Browsing



LibreOffice Doc Edit



Post Lab Questions:

1. Does manual allocation of CPU and memory violate rapid elasticity? Justify your answer.

Yes, manual allocation violates rapid elasticity.

Justification: Rapid elasticity (as defined by NIST) requires that computing resources can be scaled up or down automatically, quickly, and with minimal human intervention, often appearing unlimited to the user.

When CPU and memory are manually allocated, scaling:

- Requires human action
- Is slower and less responsive
- Cannot react instantly to workload changes

Therefore, manual allocation does not meet the “rapid” and “elastic” aspects of rapid elasticity.

2. Which NIST cloud characteristics are *not fully satisfied* by VirtualBox/KVM? Give reasons.

VirtualBox and KVM are virtualization technologies, not full cloud platforms. They do not fully satisfy several NIST characteristics:

1. Rapid Elasticity

Scaling resources usually requires manual configuration or VM restart.

No automatic, on-demand scaling.

2. Measured Service

No built-in pay-per-use metering or billing.

Resource usage is not automatically tracked for chargeback.

3. On-Demand Self-Service (partial)

Users often need administrator involvement to create or modify VMs.

Not fully self-service via a portal or API like cloud platforms.

4. Broad Network Access (partial)

Access is often limited to local networks or specific configurations.

Not inherently designed for wide, public network access.

3. Give one real-world example where virtualization alone is sufficient, and one where cloud computing is required.

Virtualization alone is sufficient Example: A company running internal development and testing environments on a single server using VirtualBox or KVM.

Reason:

- Workloads are predictable
- No need for auto-scaling or pay-per-use
- Local control is preferred

Conclusion:-

This experiment helped understand virtualization using VirtualBox/KVM. It showed how multiple virtual machines can run on a single system, improving resource usage and flexibility. Virtualization forms the basic foundation of cloud computing.